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A framework proposal for machine learning-driven agent-based models through a case study analysis

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ABSTRACT

Agent-based modeling (ABM) has been widely employed by researchers in various domains. Developing valid and useful agent-based models (ABMs) imposes challenges on the modelers. Using machine learning (ML) techniques in ABMs may facilitate the development of these models and improve their performance. This paper provides a detailed overview of the relationship between ML and ABM approaches. The benefits and drawbacks of data-driven ABMs are evaluated. A main scheme for utilizing ML techniques in ABMs is provided and explored through references to the relevant studies. As part of the primary scheme, a framework for modeling agent behaviors in ABMs utilizing ML approaches is proposed. In the framework, theoretical support is also combined with ML approaches in order to increase the accuracy of agent behavior generated by ML approaches. Using the suggested framework, a real-world case study is performed to investigate the application of ML techniques to improve the accuracy of ABMs and facilitate their creation. The findings indicate that ML approaches may facilitate the construction of ABMs.

1. Introduction

Agent-based modeling (ABM) was initially developed for the purpose of analyzing complex adaptive systems. Complex systems evolve as a result of their constituent elements and their interaction. In ABM, these elements are referred to as “agents”. Agents are defined as entities capable of making decisions [1]. ABM is primarily used to deduce the system’s behavior (macro-level behavior) or emergent behavior by modeling the agents’ individual behavior. The macro behavior of the system emerges as a result of the agents’ individual behavior (micro-level behavior) and interaction [1,2]. For example, a building’s total evacuation time depends on several micro-level factors such as how long it takes people to decide to evacuate, which exits they use, how quickly they evacuate, whether people leave alone or in groups, and how much information they have about the building’s escape plans [3]. In [4], the authors define ABM as a computational model that simulates the behaviors and interactions of autonomous agents to understand the behavior of a system. In [5], the authors define Agent-based Simulation (ABS) as a function of agents, their environment, and their interaction with other agents and the environment. In this study, ABM refers to the conceptual agent-based model, while ABS refers to its computer implementation.

With the help of agent-based models (ABMs), researchers may better understand the behavior of systems, predict how systems will behave, and address some technical challenges. ABMs are widely used in a wide variety of disciplines, including social sciences [6], health [7], ecology [8], natural sciences [9], environmental sciences [10], archaeology [11], transportation systems [12], business and management [13], and solving engineering problems [14]. Different classification schemes for ABMs have been proposed in the literature. In [15], the authors, for example, grouped the models into three categories based on how the system’s agents behave: (i)

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Table 1
Examples of ABMs created for various objectives in various application domains.

Application area	Example study	Objective
Natural Sciences	[9]	Understanding
Business and Management	[8,13]	Understanding, Understanding and Optimization
Health and social care	[7,20]	Prediction, Understanding
Engineering	[19]	Optimization
Environmental Science	[10,21]	Understanding and Optimization, Understanding
Social sciences	[14]	Prediction and Controlling
Humanities and arts	[11]	Understanding
Traffic and Transportation System	[12,18]	Prediction, Understanding

physical models, (ii) economic models, and (iii) sociological models. On the other hand, Klügl [16] classifies ABMs into three groups based on their intended use: prediction, understanding, and optimization/controlling. This classification aligns well with machine learning (ML) techniques. Based on this classification, the main scheme for the use of ML in ABMs is established (see Fig. 3). Since the suggested scheme relies on this classification, the following describes this classification.

- Prediction: It is employed in the production of forecasts for the purposes of planning operations. Using a traffic system simulation, for example, real-time bus locations can be estimated, allowing for real-time planning of bus routes [17].
- Understanding: It is used to explain observed phenomena or to compare alternative hypotheses that might account for the observed situation. In [18], the authors examined the effects of people's competitive and cooperative behaviors on crowd dynamics during evacuation and contributed to a better understanding of these behaviors' implications on crowd dynamics.
- Optimization: It is used to enhance the understanding and testing of various designs, as well as to define the proper behavioral rules of system elements in order to optimize the system's performance measure. For instance, it can be used to test various strategies for controlling an epidemic and selecting the most effective one (by the best criterion to be defined). In [19], the authors analyzed the strategic behavior of power generation companies under various market clearing mechanisms by developing an agent-based simulation model that incorporates a game-theoretic understanding of the electricity market's tender mechanism and the generation companies' learning mechanism.

ABMs can be created for more than one of these objectives, such as for both understanding and optimization. Table 1 shows examples from the literature from different application areas, grouped by these primary objectives.

When creating models, modelers frequently draw on existing theories, hypotheses, and expert opinion. For the development of an ABM, theories, in particular, are essential. Theory-driven ABMs in this study refer to models that are built on theories. Theory-driven ABMs are also referred to as traditional or classical ABMs. For the modelers, theories provide a framework but there are no solid instructions on how to use them when developing their models. Additionally, there might be multiple theories that could be applied when developing an ABM [22]. When building a model, a modeler must decide which theories to incorporate into the design. These are some of the difficulties that modelers have encountered when creating theory-driven ABMs. Some of these challenges are addressed more thoroughly in Section 2.2.

On the other hand, with the abundance of data available recently, data-driven ABMs have gained particular appeal. The data has already been widely used in the parameter calibration and model validation processes of ABS models. In the recent few decades, as individual-level data has increased, it has started to be used for the initialization of agents' parameters and simulation environments, as well as for the production of behavioral rules for agents. The term "data-driven ABMs" refers, according to the available literature, to ABMs that use data to initialize simulations and derive the behavioral rules of agents. The various uses of data inside ABMs are described in Section 2.3. For example, [12] use ABM to examine the potential benefits and downsides of taxi sharing. They use taxi data from New York City and city spatial data to create the realistic simulation environment. Some studies employ supervised ML approaches to derive the behaviors of agents in their models [13,23–25]. Using ML approaches to model an agent's decision-making process and obtaining their behavior may be beneficial, and their application in ABMs requires additional attention from the agent-based community.

Despite an increase in the number of studies employing ML in ABMs over the previous decade, the use of ML in ABMs is rarely discussed. In this study, ABMs that derive agent behaviors using ML are referred to as ML-driven ABMs. In [26], the authors explore the use of ML in ABMs by citing the relevant literature, and they suggest that it should only be used sparingly due to the costs associated with utilizing ML in ABMs. In the same vein, the relationship between ML and ABM has recently been examined in the study [27]. However, they do not investigate and test the application of ML in ABMs in a real-world scenario. To the best of our knowledge, no work has ever investigated and assessed the use of ML when deriving the agent's behavior in a real-case study.

The main objective of this study is to investigate the use of ML techniques to enhance the accuracy of ABMs and simplify their creation. First, data-driven ABMs are reviewed to have a better understanding of their current standing in the literature. Using the terms "data-driven ABMs", "ML and ABM", "reinforcement learning and ABM", and "learning and ABM", a literature search is conducted. And with the guidance of this research, the relationship between empirical data, ABM, and ML is demonstrated (see Section 2.3). After establishing the relationship between ML and ABM, it is explored how ML techniques might be utilized to improve the most challenging component of ABM's design: the derivation of rules for the behavior of individual agents.

The state-of-the-art ABM articles are studied via the viewpoint of ML in ABMs. A main scheme is presented for the utilization of ML methods in deriving agent's behaviors (see Section 3). This scheme addresses the primary purpose of the study. But every part of this scheme is not explained and investigated in this work because it would be too lengthy. As a subcomponent of the scheme, the use of supervised learning approaches for deriving agent behaviors is investigated. Lastly, a framework is proposed for the use of supervised learning techniques to the derivation of agent behavior. A theory-support is also incorporated into the framework in order to enhance the accuracy of the agent's behavior derived from supervised learning techniques. Through the suggested framework, the usefulness of ML approaches in the creation of ABMs is investigated using a real-world case study.

In summary, the purpose of this paper is threefold: (i) to discuss the challenges of the traditional ABM development process, as well as the advantages and disadvantages of data-driven ABMs, (ii) to provide a primary scheme for constructing ML-driven ABMs and a framework for ML-driven ABMs that will be developed for prediction purposes, (iii) to examine the performance and applicability of ML in ABMs through a real-world case study.

The remaining sections of the paper are structured as follows: Section 2 provides a detailed definition of classical ABMs and data-driven ABMs and discusses them. Section 3 describes the proposed scheme and sub-framework, and Section 4 applies the proposed sub-framework to a real-world case study. Section 5 discusses the limitations of the framework, and Section 6 summarizes the findings of the study and outlines future research plans.

2. Background

This section will provide background information on both the traditional ABM development approach and the data-driven ABM development approach, as well as analyze the benefits and drawbacks of each.

2.1. Classical ABM development approach

Following an assessment of various published agent-based studies, the framework represented in Fig. 1 was built to describe the traditional approach used in the building of ABS models.

When developing an ABM, a modeler performs the following steps [2,16]:

1. Search for theories about the system being modeled, where these theories are grounded on empirical evidence.
2. Construct a conceptual model utilizing these theories. This step has the following substeps:
   (a) **Categorization of agents**: The agents in the system are classified using either the real-world system or data from the literature.
   (b) **Define the interactions between agents, and agents and environment**: Ensure that environmental elements and agent interactions are in place when required. Interactions with other agents and the environment can shape an agent's behavior.
   (c) **Definition of agent behaviors**: The modeler attempts to deduce agents' coarse behavior either through observation of real-world agents or through examination of theories and hypotheses in the literature.
   (d) **Defining the agent architecture**: The modeler must decide on the agent's architecture based on the coarse behavior description, which treated the agents primarily as black boxes. It can be as simple as if-then-else rules or as complex as Belief Desire Intention (BDI) architecture. The criteria for choosing the options in this list are dependent on the complexity and flexibility of the required agent behavior.
3. The ABM model is constructed using the conceptual ABM inputs. This step involves the formulation of the behaviors of the agents. The modeler provides a step-by-step protocol for simulating an agent's behavior and interactions.
4. The conceptual model is implemented in a computer using a programming language or a specific simulation software. This step involves a verification procedure that ensures the computer model is consistent with the conceptual model.
5. The ABM parameters are calibrated using empirical data. Following the calibration step, an empirical data-based validation procedure is used. Macro-level validation is commonly used in the literature due to a dearth of micro-level data. Additionally, validated simulation models are utilized to validate generated models.

2.2. Challenges with classical ABM development process

When attempting to develop an ABM, the modeler faces various challenges. The studies all have a common set of challenges associated with ABMs [28–30]. Several studies also discuss the domain-specific challenges that come with ABMs [31]. A recurring set of challenges is inherent in the nature of ABMs. These challenges are classified and described as follows:

- **Design**: In ABM building, the first challenge is to design the model. This involves formulating the processes as rules and specifying the processes to be represented, as well as defining the level of detail that is appropriate for the purposes of the model. Defining the actual world behavior as a set of rules is the most challenging task that ABM development faces. The difficult part of the model design is the search for rules that are simple and elegant enough to summarize a fundamental theoretical question, but yet yield the behavior of interest. The simulation models are a simplified representation of reality. So they do not include all the important details but does contain only the most significant aspects. And those aspects should be represented by simple rules. Unfortunately, there is no recipe for producing ideal and simple rules.

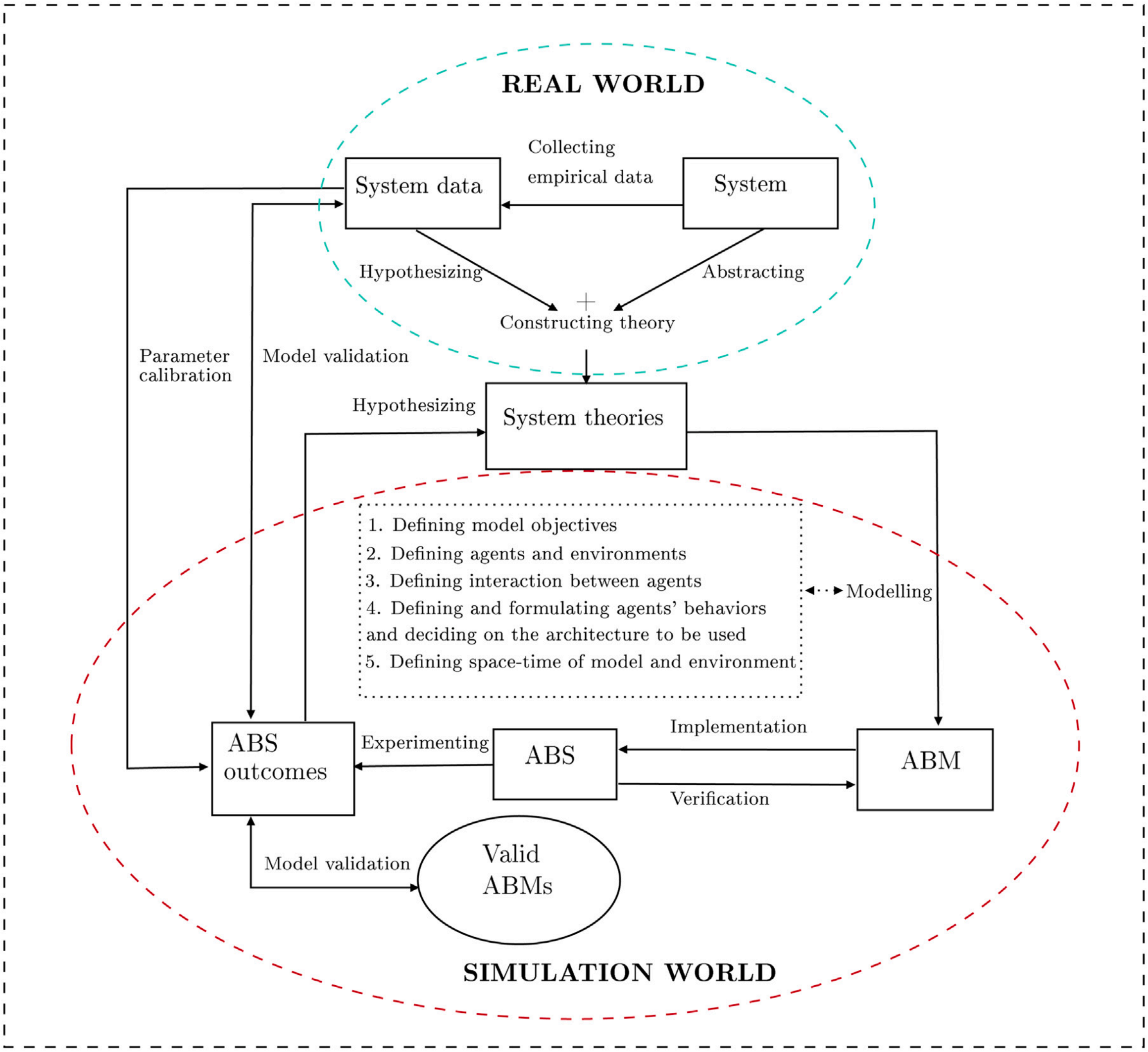

Fig. 1. Classical agent-based model developing framework.

- **Implementation**: ABMs face another challenge when it comes to developing their computer models. These difficulties might be caused by either hardware or software. On the software side, the model can be constructed either via the use of a programming language or through the use of existing simulation programs. According to their background, the modeler must select one of them to implement their model. If they are unfamiliar with computer science or are inept at using computer programs, implementing their model can be extremely taxing. On the hardware side, the generated model may be more complicated and may have a high degree of computational complexity.
- **Community**: The most frequent criticism leveled against ABMs is that they are built as one-off models. Researchers, on the other hand, suggest that ABMs should be shared and extended and reused in other studies.

This study focuses on the challenges associated with design, particularly those associated with determining the behavioral rules of agents during model design. And it investigates how data and ML approaches can help the modeler in overcoming this difficulty.

2.3. Data-driven ABMs

An agent-based model created utilizing data is referred to as a "data-driven agent-based model" in the literature [32–34]. A brief survey is conducted in this section to establish the relationship between empirical data, ML techniques, and ABM. Fig. 2 depicts their relationship with respect to the phases of ABM development. Different applications of ML in the ABM development process are highlighted by the superscripts on ML. For instance, ML^1 denotes the use of ML for preparing ABM input data. Also, Fig. 4 depicts

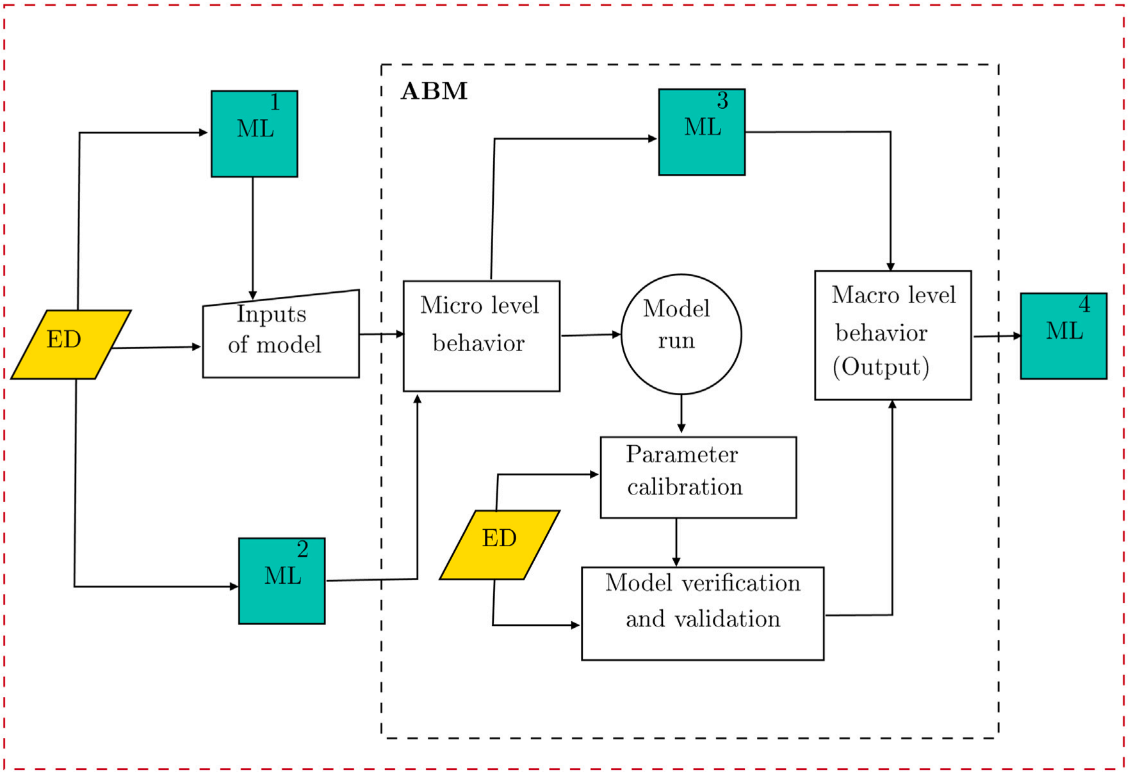

Fig. 2. ML, Empirical Data (ED) and ABM relationship.

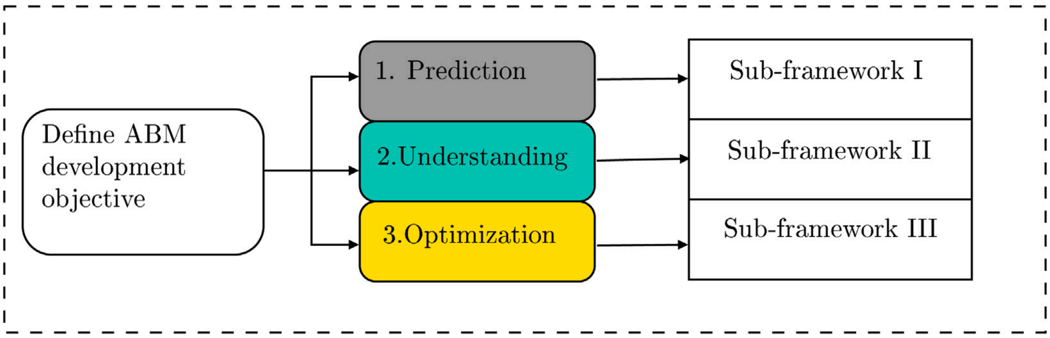

Fig. 3. Main scheme for developing ML-driven ABMs.

at which stages of the ABM development process empirical data is utilized. In a nutshell, it illustrates the ABM development process from the perspective of using machine learning and empirical data to facilitate its development.

The following cases explain the link between empirical data, ML, and ABM.

- **Case 1**: Inputs for ABMs are drawn from empirical data. Inputs can be associated with agent-specific or environment-specific attributes. For example, in [6], the authors investigate the effect of socioeconomic factors (income and education) on individuals' family formation decisions. They incorporate empirical data into their ABM to analyze family formations. In [12], the authors use ABM to examine the potential benefits and downsides of taxi sharing. They use taxi data from New York City and city spatial data to create the realistic simulation environment.
- **Case 2**: The empirical data can be preprocessed using ML models before ABM deployment so that their input data is suitable for usage with ABMs. For example, in [35], the author employs clustering and feature selection operations to process the data needed to model the decision-making process of agents in their ABM.
- **Case 3**: ML approaches can be used to infer agent behavior in ABMs. Using the appropriate data and supervised ML algorithms, agent behaviors or behavioral rules can be inferred directly. Deriving agent behavior can be viewed as a prediction problem

Table 2
ML approaches and ABM relationship.

Alternative	Supervised learning	Semisupervised learning	Unsupervised learning
ML[1] (Case 2)	✓		✓
ML[2] (Case 3)	✓	✓	
ML[3] (Case 4)	✓		
ML[4] (Case 6)	✓		✓

in ML. The agent's behavior can be inferred by acquiring the relevant attributes and then specifying the behavior of the agent as the target of the applied ML method. For example, in [23], the author creates the behavior of the agents in the classical segregation model using an artificial neural network (ANN) model. They draw the conclusion that the same outcome may be achieved with a trained neural network for deriving agent's behaviors as with manually defined behavioral rules. The author later proposes a general framework based on ANNs for generating agent behaviors in ABMs in their subsequent study [24]. Using reinforcement learning algorithms is an alternative way for obtaining agent behaviors. By configuring the environment for agents, the agents may learn the desired behavior through trial and error. For example, in [25], the authors combine reinforcement learning and decision trees to discover the agents' behaviors. They generate the agents' behaviors using a Q-learning algorithm and then use decision trees to abstract and interpret the generated agents' behaviors. Inverse Reinforcement Learning (IRL) could be utilized to derive agent behaviors. The objective of IRL approaches is to uncover the motivation underlying the preferences of agents [36]. Therefore, it may be used to learn the motivation behind the observed behavior of agents, and the modeler could then generate the agent's behavior using reinforcement learning (RL) utilizing the motivation behind the observed behavior. For example, in [37], the authors propose a method for creating an ABM using IRL. In the same vein, in [38], the authors employ IRL to model the driver behavior. They try to grasp the reward-process behind driver behavior in order to precisely imitate the decision-making process of drivers.

- **Case 4**: Supervised learning methods are used as proxy models (or meta models) for ABMs to predict the input–output relationship. There is a vast amount of work on the use of ML techniques as surrogate models for ABMs, and readers interested in learning more can read the paper [39].
- **Case 5**: Empirical data are widely utilized in ABMs to calibrate parameters of the models and validate models. In [40], the authors provide a data-driven ABM for capturing the dynamics of migration systems. They use survey data to calibrate the parameters of the equations that control the migration behavior of agents. In the same vein, in [41], the authors use data-driven ABM to assess the long-term impact of income inequality on consumption. Data is used to calibrate each agent behavior parameter in their model, but they also directly extract some agent behaviors from historical data.
- **Case 6**: ML techniques can be used to turn the simulation model's output into a format that is useful for decision making. In [42], the authors attempt to solve the supplier selection problem by applying supervised ML methods to the make-to-order simulation model's output.

Table 2 classifies ML approaches based on how they relate to ABMs. It shows, for each case, which machine learning approaches can be employed. It states, for instance, that if a modeler encounters Case 2, they may employ supervised learning and/or unsupervised learning techniques. Alternatives in Table 2 refer to Fig. 2. This study focuses on Case 3, and the subsequent section of the paper delves into that case in detail and proposes a framework for it.

2.4. Building data-driven ABMs

This section will discuss the benefits and drawbacks of developing data-driven ABMs. The advantages of building data-driven ABMs can be listed as follows:

- **Better representation of real systems**: Real data from the system aids the modeler in developing a more accurate representation of the system. For instance, the data-driven agent behaviors can capture the real behaviors of the system. For example, in [43], the authors propose a data-driven model for a more accurate representation of the evacuation scenario's dynamic environment. Good data also might represent the observed behavior with all transparency. It eliminates the modeler influence, allowing the modeler to create a more pure model.
- **Ease of the modeling process**: When considering Cases 1 and 3 from the preceding section, a data-driven approach may make the modeling process simpler. In Case 1, the modeler could quickly set the simulation environment and agent's attributes by utilizing the data for model initialization. In Case 3, the formulation of agent behaviors in the model is one of the most challenging aspects of designing an ABM (see Section 2.2). This challenge stems not just from the difficulty of translating theories to agent behaviors, but also from the possibility that no theory or established rules of behavior exist. In addition, there may be several theories, and modelers may be unable to choose which theories to incorporate into their model. In [44], the authors underline the variety of theories on human behavior and the difficulties in translating these theories into models of human behavior. They give a framework for mapping the six most prevalent behavioral theories (the following theories are given: rational actor, bounded rationality, prospect theory, theory of planned behavior, descriptive norm, habitual/reinforcement learning) to human behavior to overcome these challenges. In [45], the authors compare the performance of expected utility theory and an extended version of prospect theory in ABMs of financial markets. The results

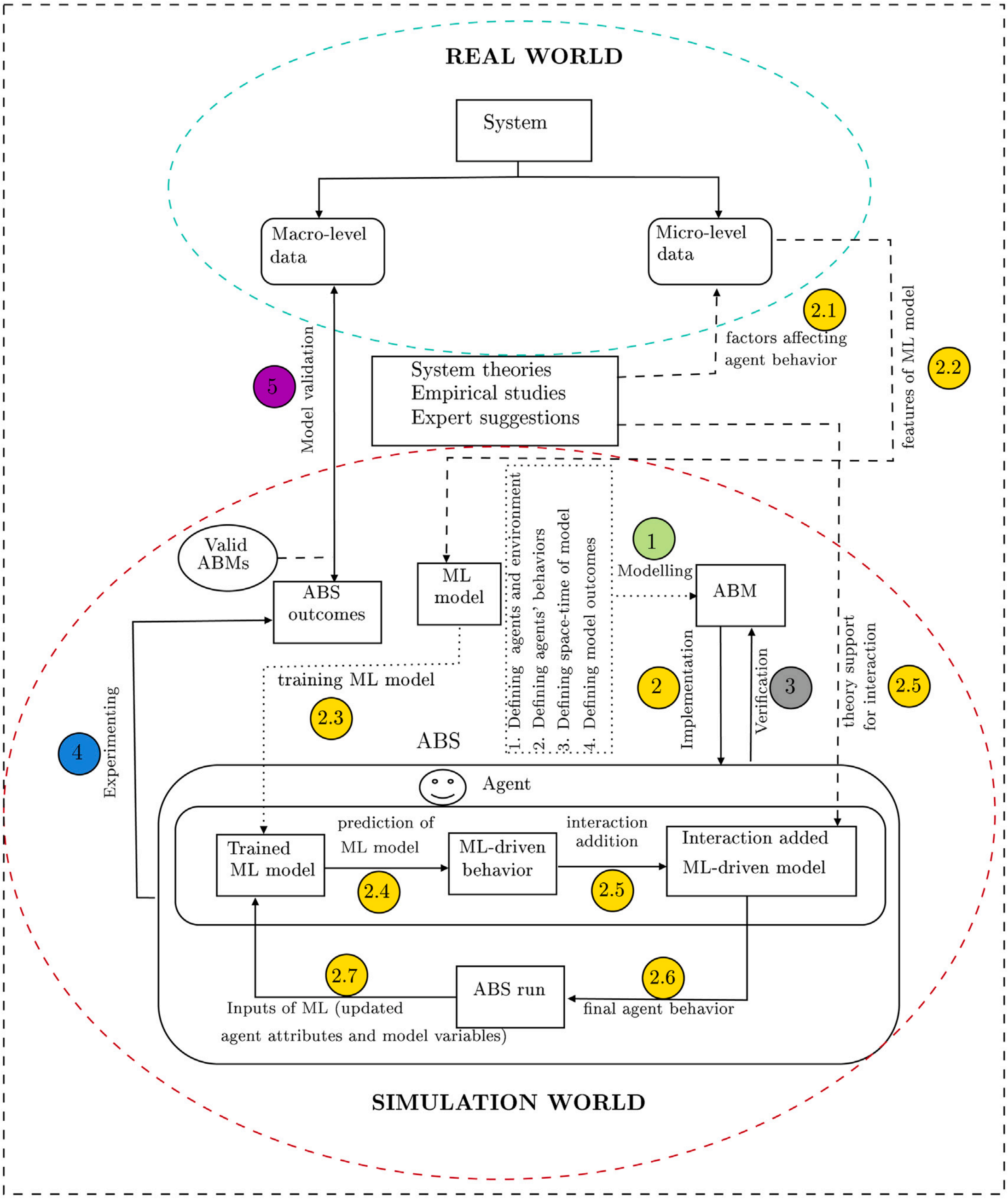

Fig. 4. Sub-framework I.

show that prospect theory is superior to expected utility theory when it comes to the success of agents in the market. Data-driven methods also facilitate calibration of ABMs by eliminating the need for tedious parameter fine-tuning [46]. Considering all these problems, the modeling process may be simplified by extracting agents' behaviors using data-driven solutions.

- **Utilizing current data resources**: The amount of data that is currently accessible is quite staggering [47]. This data could be utilized by practitioners to construct ABMs and enhance the usefulness of the stored data sets. Companies, for instance, generate large amounts of customer data daily and store them in their databases [48]. They could utilize this data to create an ABM model. By establishing such an ABM model, companies may gain a better understanding of customer expectations, test the impact of their policies on customers, and create more effective strategies to meet customer demand.

Data-driven ABMs, on the other hand, suffer from a number of drawbacks. These drawbacks are as follows:

- **Availability of data**: Despite recent advancements mentioned in the study [49], there are no efficient methods for directly measuring human behavior patterns [50]. Researchers often use social surveys to collect indirect data on human behavior, such as interview questions [51–53]. Therefore, additional work is required to derive behavioral rules that emerge from indirect measurements. If there are insufficient behavioral data, the behavioral rules derived from the data will be invalid, and the macro-level results of the ABM model will be incorrect. So, data collecting is essential for developing the behavioral rules of agents. In [54], the author reviews the existing data collection techniques used in ABMs. And he explores how mobile information and communication technology can assist in enhancing the decision-making process of agents in ABMs.
- **Generalization of the model**: Deriving agent behaviors using supervised learning approaches may result in scenario-specific agent behaviors, i.e., the modeler is unable to obtain generic agent behaviors [55,56]. For instance, if all observational data are gathered in the summer but inferred behaviors are used to simulate agent behavior throughout the winter, the findings may be erroneous. This restricts the applicability of the generated ABM and decreases the generalizability of the ABM. In addition, ABMs might be used to construct theories in the social sciences [57]. So, if the ABM is built for such expansive aims, constructing models based on data (especially inadequate and biased data) will not result in the creation of credible theories.
- **Interpretability of model results**: The model is said to be interpretable if the modeler has an understanding as to why the model's outputs occur [55,58]. In other words, the modeler wants to grasp the connection between input and output. If the modeler creates ABMs with black-box ML models, it will be difficult to interpret the model output in a transparent manner. If the modeler needs an interpretable model, they must employ white-box ML models [59].

3. Main scheme for ML-driven agent-based simulation

A main scheme is built for ML-driven ABMs based on the model's development objective, as shown in Fig. 3. It is designed specifically to establish how ML techniques may be utilized to derive agent behaviors. This scheme serves as the foundation for the authors' more in-depth investigation of ML-driven ABMs. The rationale behind the structure of the scheme is explained by referencing relevant studies.

This proposed scheme concentrates on how to derive agent behavior using ML techniques (see Case 3 in Section 2.3). Based on the findings of the literature, it can be stated that there might be two approaches for acquiring the agent's behavior in data-driven ABM: (i) obtaining agent behaviors directly through the use of supervised learning techniques (ii) attempting to ascertain the underlying causes of observed behaviors and then reproducing observed behaviors through the use of statistical techniques. A researcher must select the one that best serves the purposes of an agent-based model. So, it would be beneficial to discuss the relative merits and demerits of the two approaches.

- It may be simple and fast for modelers to obtain agent behaviors by using supervised learning techniques. If an ABM is used for prediction, this approach has the potential to produce accurate and satisfactory results.
- However, if ABM is used for understanding (i.e., ABM operates in a white-box fashion), they may fail to accomplish this, as some supervised learning techniques (deep learning, support vector machines, and so on) operate in a black-box fashion, i.e., they do not provide an explanation for the underlying causes of the behaviors. According to [60], black box models usually have limited use in the social sciences because it is difficult to collect further information on human processes without the ability to show how the input variables are connected to the output. Given the widespread use of ABM in the social sciences, it appears that models obtained through supervised learning are unlikely to be accepted.
- IRL has enormous potential for elucidating the underlying causes of agent behaviors. It can be used to infer the motivations behind agent behaviors, particularly when developing ABMs for understanding. Then, these agents' motivations can be translated into simpler rules describing their behaviors. However, IRL model building requires sequential behavioral data from agents. Fortunately, there is a vast amount of human trace data available today, such as social network and mobile phone data.
- If ABMs are build for controlling, i.e., obtaining optimal behaviors for policy making, reinforcement learning algorithms may be used. Reinforcement learning algorithms enable the modeler to create ABMs with intelligent and rational agents. In other words, RL enables the modeler to obtain optimal agent behavior in their ABMs if the simulation environment is stationary [61]. In [62], the authors use reinforcement learning algorithms in a BDI framework to simulate the evacuation behavior of humans.

Table 3 highlights the connection between model development objective, employed ML approach, and agent behavior type modeled. For instance, if modelers want to use the developed model for controlling, reinforcement learning techniques can be used to obtain the optimal behavior (a behavior type of agent) of agents. This study will not cover all of the sub-framework of the scheme because each sub-framework should be the subject of a separate paper. The rest of this study is to describe Sub-framework I in detail and to illustrate it through a comprehensive and detailed real-world application.

3.1. Sub-framework I

Using supervised-learning approaches to simulate agent behaviors in models may raise various concerns that must be addressed if these techniques are to be used. In the framework, these challenges are addressed, and a solution is offered for each one. These questions are as follows:

Table 3
Learning approaches for extracting agent behaviors.

Learning approach	Prediction	Understanding	Controlling	Behavior type
Supervised learning	✓			Human behavior
Reinforcement learning		✓	✓	Optimal behavior
Inverse reinforcement learning		✓		Human behavior/Optimal behavior

- How can one determine which data may be utilized in an ABM to infer agent behavior using ML? It is suggested to conduct a literature search to determine the factors influencing the modeled behavior, followed by data collecting based on the literature search results. This will increase the interpretability of obtained model.
- Which ML models should be implemented? If modelers also consider the interpretability of their model, it is recommended to use tree-based ML techniques because they are white-box models. Alternatively, if a modeler only wants high prediction accuracy from their ABMs, they can employ alternative supervised learning techniques, such as support vector machines, deep learning, and so on.
- How to model the interaction between agents if the agent's behavior is modeled directly with ML models? The data used to train an ML model might not contain any variables relating to the interaction between agents or agents and the simulation environment. In such a situation, the constructed model might not even be referred to as an ABM, and its results may not be desirable. It is suggested to combine the output of ML with the interaction element in order to obtain agent behavior in the model. Before adding the interaction component to the output of ML, the modeler must conduct research into the relevant literature to obtain and understand the hypotheses and theories that explain the interactions between agents.

The findings of the literature review also indicate that combining theories and expert advice with ML approaches could be valuable for generating more robust and effective models. In [63], the authors discuss the difficulties of determining the behavioral rules of agents. They use a combination of expert knowledge and ML methods to learn the behavioral rules of the agents. In [60], the author discusses the advantages and limitations of both data-driven and theory-driven ABMs in terms of model quality and model construction in social sciences. As a result, the author presents the TIMMIT concept (which stands for Theory-Interpretable Models and Model-Interpretable Theory). The principle states that all social science models incorporating large data should be theoretically interpretable (not black boxes), and all social science theories that employed in the models should be model-interpretable (falsifiable). In [64], the authors build a data-driven ABM that employs ML for the criminal decisions of agents. Important difference between their studies and other data-driven investigations is that they strongly support their model outcomes with theoretical studies. This makes their model more grounded and their model findings more explicable. In [65], the authors compare the theory-driven and empirical-driven ABMs. When designing their empirical ABM, they replace the theoretical assumptions of theory-driven models with empirical data findings. They employ sensitivity analysis to compare the robustness of two approaches. The findings indicate that theory-driven models are more tolerant to model agents' behavior changes.

In addition to the aforementioned difficulties, there are a number of additional questions concerning the performance of constructing an ML-driven ABM. How the prediction accuracy of individual-level behavior of agents (obtained by trained ML models) would affect the outcome accuracy of ABMs at the macro-level? How to link the dynamic nature of a simulation environment with the static outcomes of an ML model? These concerns are addressed and analyzed in the case study in light of the developed model's results.

Consequently, based on the preceding discussion and evidences from the literature, a framework for creating a theory-supported ML-driven ABM is presented. Fig. 4 portrays the proposed framework by emphasizing the distinctions between theory-supported ML-driven ABM development and classical agent-based model building processes. The application steps of the framework when constructing an ABM are denoted by the circled numbers in Fig. 4 and are summarized as follows:

1. The modeling phase is the initial step of the framework. Modelers design a conceptual model in a similar manner with the classical ABM development methodology at this stage. However, they are not required to formulate the behavior of agents or specify their architectures in order to represent their behavior.
2. The second phase of the framework, "model implementation", is the framework's central component and consists of a series of substeps. In this phase, the modeler addresses the most challenging aspect of ABM design: specifying agent behaviors. For this purpose, the following sub-steps are employed:

   2.1. The modeler does a literature search to identify the variables that influence the behavior of the modeled agents. They may utilize system theories, empirical data, and expert recommendations for this aim. Following this search, they determine what micro-level data they will require and collect the necessary data.

   2.2. At this stage, the modeler derives the features of the ML model. The modeler performs all data preparation stages for the development of an ML model [66].

   2.3. The modeler trains an ML model using the previously stated features. Agent behavior will be the output of the ML model. They apply all traditional processes of training a supervised ML algorithm [66]. If interpretation is required, the modeler may use white-box supervised learning algorithms; otherwise, black-box supervised learning methods may be preferred.

2.4. The modeler obtains the agent's behavior by feeding the agent's attributes and other simulation environment data to a trained ML model.
2.5. The next step is to incorporate agent interaction into the ML model's output. For this, modelers must examine the relevant literature. They may employ system theories, empirical studies, or expert guidance.
2.6. After acquiring the necessary information on the effect of agent interactions on agent behaviors, the modeler must find a procedure for systematically incorporating the interaction into the ML model's output. They may simply set some rules or formulate some formulas based on the data obtained from the literature. So, they acquire the final agent behavior that will be utilized in ABS.
2.7. Throughout the ABS run, the updated agent attributes and simulation environment variables are fed into the trained ML model to predict the agent behavior at next simulation step.

Because Steps 3 through 5 of the framework (as depicted by the colored circles in Fig. 4) are carried out similarly to the traditional ABM development approach, they are not discussed further.

On the other hand, the differences between the proposed framework and the classical ABM approach can be summarized as follows:

- The proposed framework suggests using ML models to model agents' behaviors rather than the modeler formulating equations or defining rules as in the classical approach. This expedites the modeling process and lessens the modeler's workload and the modeler's impact on the model's quality.
- As shown in Fig. 1, the classical approach makes use of theories, empirical studies, and expert suggestions in the modeling stages of ABM, particularly in the formulation of agents' behaviors. On the other hand, the proposed framework seeks the use of them for two objectives: (i) Identifying factors that affect the behavior of the agent. Before collecting micro-level data for training an ML model, the modeler should conduct a literature search or consult with experts in order to gather the required data for developing an ML model with high prediction accuracy, (ii) Incorporating agent-to-agent interactions into ML prediction models. It is essential to include the interaction between agents in the model. If there are no features in the data set that explicitly reflect the interaction between agents, the modeler must examine the literature and add the interaction to the output of ML.

4. The case study: Migration flow in Burkina Faso

Migration flow in Burkina Faso is selected as a case study to illustrate the study's framework. In light of the proposed framework, we will build an agent-based model to predict Burkina Faso's migration flow. Before describing the model's specifics, we describe the dataset that will be employed in its development.

4.1. Dataset description

To do this study, we used a set of data from the research project called "Migration Dynamics, Urban Integration, and Environmental Survey of Burkina Faso" (EMIUB is the acronym for The Enquete Migration, Insertion Urbaine et Environnement au Burkina Faso) [67]. In 2000, the EMIUB performed a nationwide study with 3500 households and over 8000 respondents. The data provides migratory flow details pertaining to the locations of residence, work activities, marriages, and children of respondents. This data was previously used in another article to construct a theory-driven ABM [68]. The authors attempted to validate the model by forecasting the migration patterns of 4449 people who lived between 1970 and 1999. They initialize the agent-based model with the survey data collected from these individuals. We used the same procedure to compare the proposed model's performance against that of a theory-driven ABM.

4.2. Model

Here, we provide an ABM to simulate the migration behavior of Burkinabe individuals. We have employed ML models as a meta model to model the decision-making process of agents throughout the simulation. First, we will explain the ABM, then we will explain ML components and demonstrate their interaction through simulations.

4.2.1. ABM

This section corresponds to the initial stage of the framework. All conceptual information necessary for the execution of ABM is provided. We have explained the model by using The Overview, Design concepts and Details (ODD) protocol [69] since it enables a rigorous explanation of the modeling part.

- **Purpose**: The purpose of the model is to generate a flow of migrants in Burkina Faso. Agent decision-making is guided by an ML algorithm trained on migration survey data. A single grid represents the spatial resolution of the migratory flow. The temporal resolution is one year, implying that agent migration decisions are made just once every year.

Table 4
State variables.

Variable	Explanation	Type
1. Zone	Birthplace	Fixed-static
2. Sex	Gender	Fixed-static
3. Age	Age	Deterministic-dynamic
4. MaritalStatus	Marital status	Stochastic-dynamic
5. EmploymentStatus	Employment status	Stochastic-dynamic
6. Education	Education level	Deterministic-dynamic
7. MigrationExperience	Number of previous migrations	Stochastic-dynamic
8. ElapsedTime	Elapsed time from last migration	Stochastic-dynamic

- **State variables and scale**: The model consists of a single agent, an individual. Table 4 lists and describes its state variables. Demographic attributes are used to represent the heterogeneity of agents. Additionally, the agents keep information about their migrations throughout the simulation. Dynamic variables are those that are time-dependent and are updated throughout the simulation. For example, we obtained a person's marriage and divorce probability at each age, as well as updated the agent's marital status throughout the simulation. To update such variables, we evaluated the data set and calculated their probability of change throughout simulation. The model replicates a time period of 30 years using a one-year time step. The model's spatial extent is the eight zones of Burkina Faso, which are represented by a single grid. A total of 4449 agents are included in the model.
- **Process overview and scheduling**: The model is composed of three components: initialization, ML, and simulation. Persons and a single grid environment are created during the initialization process. The data set is used to extract the attributes of the agents (see Section 4.1). Throughout the simulation part, individuals decide whether to migrate to a new location or remain in their current location at each simulation iteration. After each iteration, each agent's dynamic variables are updated. The total number of migrated persons in each iteration (year) is gathered and recorded into an output file at the end of the simulation. Individual migration decisions are made by the model's ML component (see details in Section 4.2.2).

- **Basic principles**: The model complies to the fundamental principles of migration.
- **Emergence**: The annual migration flux emerges from the individual migration decisions made by each agent via their ML-driven decision-making module.
- **Sensing**: Persons are aware of their surroundings. They know where the country's zones are.
- **Interaction**: Throughout the simulation, each agent interacts with other agents in their immediate neighborhood. Each agent's migration decision is influenced by the migration decisions of their neighbors.
- **Stochasticity**: The model incorporates stochasticity in a variety of ways. Each agent in each zone is assigned a cell at random. Each iteration, the stochastic variables are updated randomly using defined probabilities and random functions. Deterministic variables are computationally updated.
- **Collectives**: Individuals belonging to the same demographic (the same sexes, ages, education levels, etc.) group exhibit similar migration behaviors. But they never behave as a group.
- **Observation**: To validate the model, the total migration flux is observed and plotted during each iteration of simulation.
- **Learning and Adaptation**: As the migration experience is fed into the ML model, it has an effect on the agent decision-making process iteratively. In this way, it can enable the agent change their migration behavior throughout the simulation implicitly.
- **Objectives**: Agents have no explicit objectives in the model.

Following dynamic events occur throughout the simulation:

- An agent is born in each of these regions at a birth rate of 0.083 and dies at a death rate of 0.90.
- At each simulation iteration, the marital status of each agent is updated based on the probability function generated from the EMIUB data set.
- Throughout the simulation, the agents get older.
- After each movement, an agent gains more experience in migration.
- Throughout the simulation, an agent's education level is shaped probabilistically based on the data set.
- We assume that an agent's employment status at each year follows the Markovian property and obtained transition probabilities by analyzing data. The employment status of an agent is updated throughout the simulation based on the computed probabilities.
- Throughout the simulation, the elapsed time (difference between the latest migration time and the current simulation time) is updated in accordance with the agent's migration decision.

Up to this point, we have explained the fundamental concepts of the ABM; in the next section, we will describe how the agent's decision-making process is implemented using ML models.

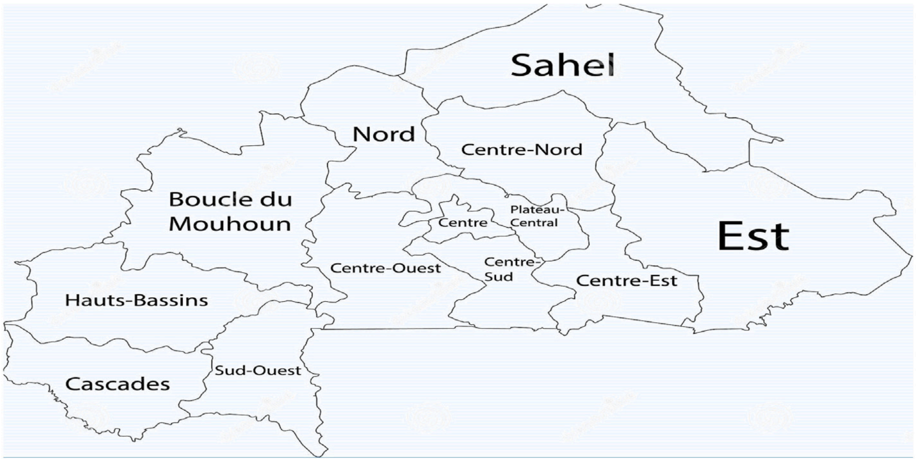

Fig. 5. Map of Burkina Faso: Main regions of the country.

4.2.2. ML model

Burkina Faso is divided into thirteen administrative regions, as shown in Fig. 5. In the dataset, some of these locations are combined under a single name, and then the birthplaces of individuals are divided into eight regions. At the start of the simulation, agents are placed in one of these regions, and they move between these regions or other countries during the simulation. The modeler needs address the following questions in order to model this migration behavior of agents: (1) How does an agent make migration decisions, (2) What destination does a migration agent choose. The answers to these questions represent the migration patterns of the agents and have a direct impact on the system's macro-level behavior. Using ML techniques and data analysis, these questions are answered, and the migratory behavior of an agent is discovered. Using the proposed framework as a reference, we will discuss in detail how we constructed the model.

- **Preparing the micro-level data**: Here, by using Steps 2.1 and 2.2 of the framework, the features of ML models are obtained. Initially, we investigate the EMIUB data set for agent-level characteristics. EMIUB data provides comprehensive migratory flow information regarding respondents' places of residence, employment status, marital status, and their children, as well as their basic demographic characteristics. Before extracting the features of ML model for predicting agent behavior, we have searched the literature on migration behavior for determining the factors affecting migration behavior of agents.
  Studies have shown that fundamental factors such as marriage, divorce, education, gender, employment status, and age have a substantial impact on the migration decisions of individuals [22,70–72]. For example, a young single female is more likely to migrate than an older married female. Individuals are also affected by the moving behavior of those around them, including family, friends and peers. Some of the other factors affecting people's migration decisions are global environmental changes [68], income and national policies [73]. Incorporating these elements can improve the accuracy and explainability of model outputs when simulating the migration decisions of agents. Individuals' marital status, age, sex, education level, employment status, and birth place are all included in the EMIUB dataset. These attributes can be utilized to train an ML algorithm to predict an agent's migration choice. For the purpose of adding the effect of migration experience and migration barriers, we have additionally constructed two additional variables for training an ML model: migration experience and time between consecutive migrations. Table 4 lists all of the inputs used by the ML models to predict agents' migration decisions.
- **Building ML models for agent behavior**: This part refers to Steps 2.3 and 2.4 of the framework. We have utilized the traditional steps of ML to predict the migration decisions of agents. These steps are as follows:

  1. **Data preparation**: Throughout the period from 1970 to 2000, individual migratory patterns and the values of other dynamics variables are implicitly included in the dataset. So, the defined input variables in Table 4 are retrieved and organized annually. For further clarification, for each individual, we have extracted its migration decision, age, sex, marital status, employment status, education level, migration time, and elapsed time from previous migration for each year from 1970 to 2000. Then, we sampled from each year's data to obtain the final data that will be used to train ML models. We have tried to choose data entry for each individual equally in order to include the diversity of individuals in the ML model. In addition, our ML challenge here is a classification problem. It will predict whether or not an agent will migrate. The output variable of the ML model is hence binary. The data set must be balanced in terms of output data for ML findings to be dependable. Thus, we have balanced the data, i.e., the number of individuals who desire to migrate and those who do not prefer to move are equal in the final data set.
  2. **Choosing a model**: We selected three tree-based models – Decision Tree, Random Forest, and the XGBoost algorithm – because their predictive performance is strong. They are also interpretable models but in this study we do not use the interpretability ability of this models since we just aims to apply the proposed framework and assess its performance and applicability in a real-case study.

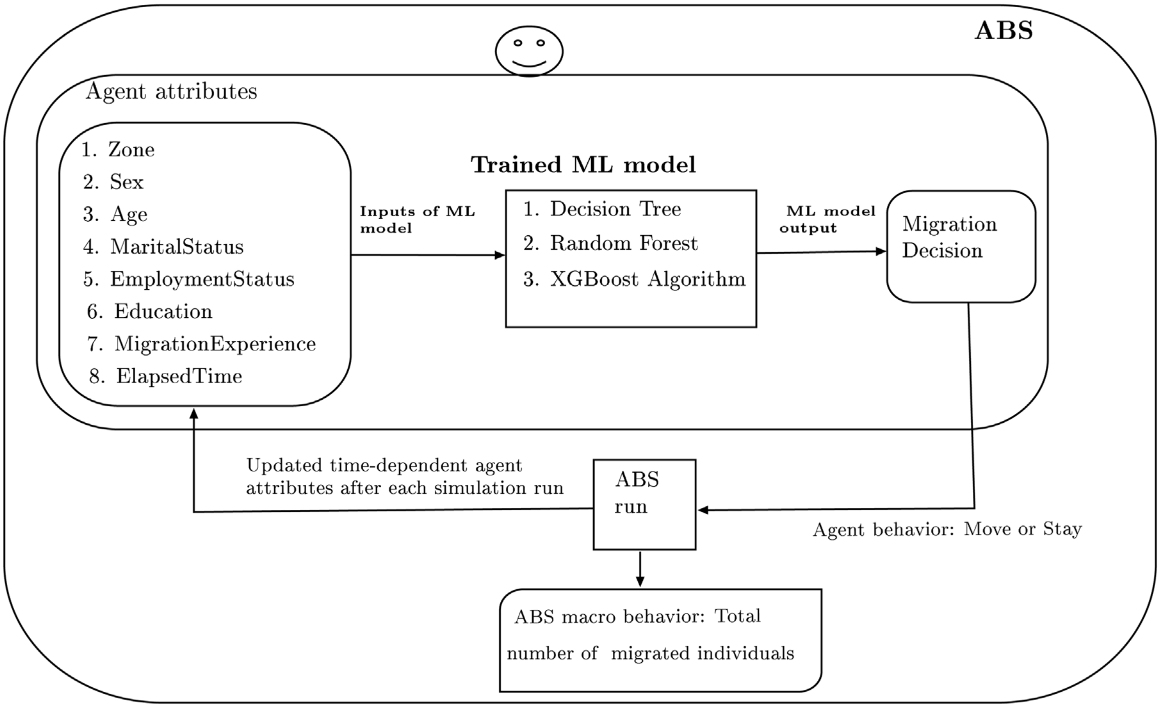

Fig. 6. The ABM workflow.

3. **Training the model**: Prior to training each model, we used grid search to determine their hyperparameter values. As recommended by the ML community, we used 80% of the data for training and 20% for testing. After that, we trained each model and obtained accuracy in both training and testing (see Table 5).
4. **Evaluating the model**: We have examined every ML algorithm for overfitting and underfitting. No overfitting or underfitting issues have been found in the models we tested.
5. **Making predictions**: In the simulation, we can utilize any model that has been trained to make migration decisions.

Fig. 6 depicts the ABM workflow and its interaction with ML models. Once the ML model has generated a decision on whether or not to move, it should be determined where the agents move. The following transition matrices, derived from migration data, were used for this purpose.

$$\mathbf{P_{urban}} = \begin{array}{r} \\ Home \\ Urban \\ Rural \\ Abroad \end{array} \begin{array}{c} \begin{array}{cccc} Home & Urban & Rural & Abroad \end{array} \\ \left\| \begin{array}{cccc} - & 0.211 & 0.364 & 0.425 \\ 0.803 & - & 0.131 & 0.0660 \\ 0.702 & 0.111 & 0.129 & 0.058 \\ 0.925 & 0.064 & 0.011 & - \end{array} \right\| \end{array}$$

$$\mathbf{P_{rural}} = \begin{array}{r} \\ Home \\ Urban \\ Rural \\ Abroad \end{array} \begin{array}{c} \begin{array}{cccc} Home & Urban & Rural & Abroad \end{array} \\ \left\| \begin{array}{cccc} - & 0.378 & 0.092 & 0.530 \\ 0.312 & 0.248 & 0.184 & 0.256 \\ 0.296 & 0.398 & 0.096 & 0.210 \\ 0.756 & 0.193 & 0.0510 & - \end{array} \right\| \end{array}$$

where P_{urban} and P_{rural} represent the urban and rural origins of the agent, respectively. The agent's current location is divided into four categories: Home (the place where the agent's family lives), Urban, Rural, and Abroad. An agent that decides to migrate moves to one of these places based on the stated probabilities.

4.3. Implementation of ABM

The model is implemented in Python using Mesa library [74]. To assess the performance of ML models in predicting the macro-level behavior of ABS, we created three ML models with a different number of features. We performed feature engineering and extracted variables 7 and 8 (see Table 4) to improve the prediction accuracy of the ML models. Model 3 is the final model and

Table 5
Performance comparison of models.

ML algorithm	Accuracy (%)	Model 1	Model 2	Model 3
Decision Tree	Training	68.3	83.5	92.5
	Testing	67.1	82.9	91.9
Random Forest	Training	69.8	83.6	93.1
	Testing	69.5	83.1	92.7
XGBoost	Training	70.9	84.2	93.5
	Testing	69.9	83.7	93.1

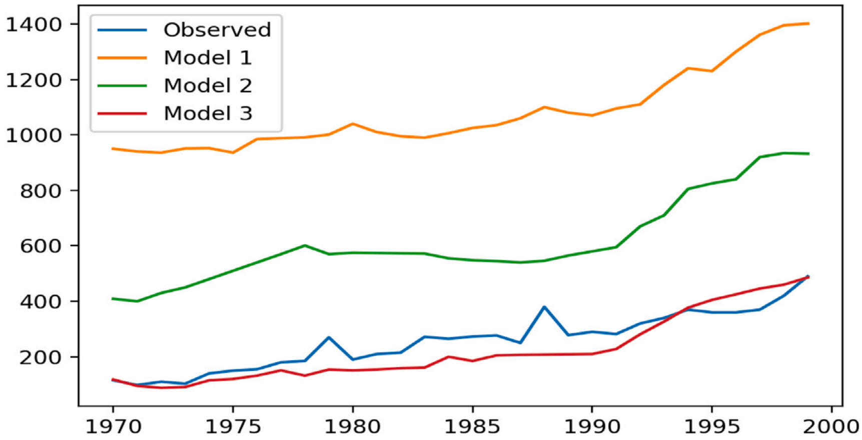

Fig. 7. Comparison of ML models performance.

contains all of the agent features specified in Fig. 6. On the other hand, Model 1 excludes both agent attribute 7 and 8 from the ML model, whereas Model 2 excludes only agent attribute 8. We obtained agent behaviors using three prominent and well-performing tree-based ML algorithms. They have similar prediction accuracies, but XGBoost outperforms the other two algorithms slightly (see Table 5). As a result, we used the XGBoost algorithm as a meta model to predict agent behavior. Fig. 7 depicts the observed total migration flux of 4449 simulated individuals from 1970 to 1999, as well as the ABS model, which employs Model 1, Model 2, and Model 3 to predict agent behavior in simulation. Despite their fairly high ML performance, Models 1 and 2 have a considerable inaccuracy in estimating overall migratory flow. There are more people migrating in Model 1 and Model 2 because they do not include the migration experiences of the agents and regard each movement as an independent one because they do not incorporate the elapsed time between each migration.

Model 3 has the best Mean Absolute Percentage Error (MAPE) value of 25.7%. It has an adequate MAPE value, but it can be enhanced. When we examine the graphs of the total migration flux for Model 3, the total migration flux line is generally below the observed values for the anticipated years. Previously, we discussed how interactions between agents can influence a person's decision to migrate. However, there is no feature in the data set that represents this influence while training an ML model. In the following section, we have integrated ML output with interactions between agents to increase the macro-level accuracy of the agent-based model, i.e., to reduce the error between observed migration flow and simulated migration flow.

4.4. Combining interaction factor with ML output

This section and Section 4.5 correspond to Steps 2.5, 2.6, and 2.7 of the framework. The empirical studies are utilized to determine how interaction influences the migration behavior of agents, and Eq. (1) is developed to incorporate the interaction into the ML output. Agents' migratory decisions are heavily influenced by their interactions with one another. Migration of neighbors and peers enhances the likelihood of individuals migrating, according to studies [75,76]. Also, without an interaction element, a model cannot be considered agent-based. Using an agent-based model is powerful since it incorporates the interaction aspect. Unfortunately, there is no feature in our ML models that takes into account the impact of interactions on migration decisions.

For simplicity, we have assumed the Moore neighborhood as an agent social network, and at each simulation iteration, an agent's migration decision is influenced by the number of migrated agents in its neighborhood during the previous simulation iteration. Additionally, it is known that individuals are more likely to be influenced by the migratory decisions of their peers. Among the agents in the neighbors of an agent, agents in the same age group as themself were assumed as the peers of this agent. The following age groups are defined: [18, 29], [30, 39], [40, 49], [50, 59], [60, 69], [70, 79].

We have utilized a probabilistic approach to describe the effect of an agent's neighbors' previous migration decisions on its own migration decision. So, we have constructed Eq. (1) for computing the migration probability of individuals. To conform to the

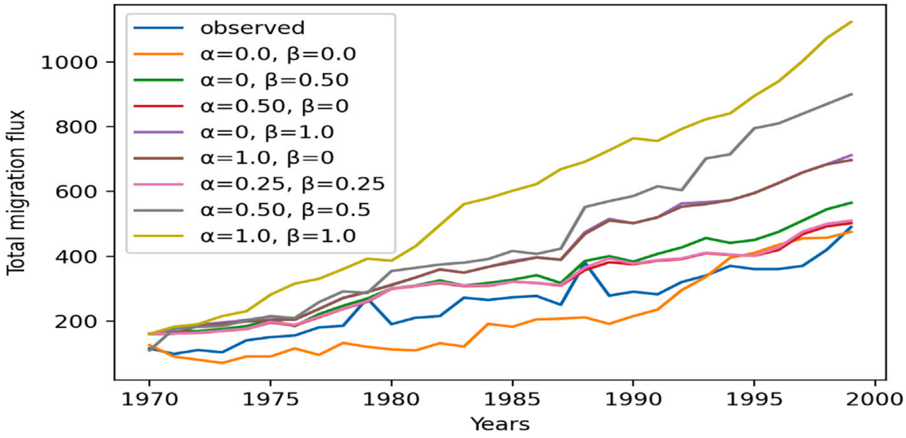

Fig. 8. Sensitivity analysis for α and β parameters. The change of the migration flow over time, with varying α and β values.

fundamental rule of probability, we have defined it in piecewise form.

$$P(migration) = \{ \begin{array}{ll} P(M_i)(1+\alpha_i(NP_i^M/NP_i)+\beta_i(P_i^M/P_i)) & P(M_i)(1+\alpha_i(NP_i^M/NP_i)+\beta_i(P_i^M/P_i)) \le 1 \\ 1 & P(M_i)(1+\alpha_i(NP_i^M/NP_i)+\beta_i(P_i^M/P_i)) > 1 \end{array} \tag{1}$$

where $P(M_i)$ is the migration probability for agent i based on the ML output. NP_i^M represents the number of non-peer migrants in agent i's neighborhood, while NP_i represents the total number of non-peer agents in its neighborhood. α_i represents the sensitivity of agent i to visual effect, while β_i represents its sensitivity to peer effect. P_i^M represents the number of migrated peer agents, while P_i represents the total number of peer agents in its neighborhood. On the other hand, determining the sensitivity of each agent to visual affect and peer affect is a challenge. We have therefore generalized these parameters ($\alpha = \alpha_1 = \alpha_2 = \cdots = \alpha_n$; $\beta = \beta_1 = \beta_2 = \cdots = \beta_n$; n is the number of agents.).

Prior to executing the simulation experiments, we analyzed Eq. (1) to determine the effect of α and β on the macro-level outcomes of the model. The model prediction accuracy can be improved by finding optimal or suboptimal values for these parameters. Hyperparameter optimization techniques from the ML domain or some heuristic algorithms or metaheuristic algorithms can be utilized to discover the optimal or near-optimal values of these parameters.

4.5. Model calibration

Finding the optimum possible model parameter settings based on observed data is known as calibration. Calibration of simple models with few parameters can be done with a pure parameter scan, but it takes a lot of computing power to calibrate complex models. The simulation optimization method and the Bayesian method are the two procedures that can be utilized for calibrating models [77]. The simulation optimization method tries to find model parameters that minimize the difference between simulation and observed output. The Bayesian technique likewise seeks to reduce the error between the simulation and observed output. But it also takes prior information regarding model parameters into consideration. It considers the unknown model parameters to be random variables with a probability distribution. ML techniques can also be used to calibrate models. The relationship between model inputs and outputs is determined using ML techniques as a surrogate or meta model [78]. The simulation optimization method was utilized in this study since it can identify model parameters faster and the model only has a few parameters. In the model, we attempt to identify the model parameters that minimize the absolute difference between the number of migrants, which is the simulation's macro output, and the actual number of migrants observed.

$$min\ \ f(t) = |\sum_t S_t - O_t| \tag{2}$$

The objective function is denoted by the Eq. (2), where S_t is the number of migrants in simulation at year t and O_t is the number of observed migrants at year t. We have no analytic form for the objective function in this situation, and the value generated by the simulation model sets its value. The function we are attempting to optimize can therefore be viewed as a black-box. So one can use the Nelder–Mead Direct search method [79], meta-heuristic algorithms [80], and hyperparameter optimization methods (e.g., grid search method, random search, etc.) to determine the optimal or near optimal values of the parameters. We utilized the grid search method to get the optimal values for α and β because we only have two parameters and the grid search method is simple to implement.

The values of the parameters (α and β) change from zero to one. The optimal values may be any combination of these two parameters. Therefore, it is nearly impossible to perform grid search to all continuous values of these parameters. We have employed multi-step calibration with the objective of minimizing the MAPE value.

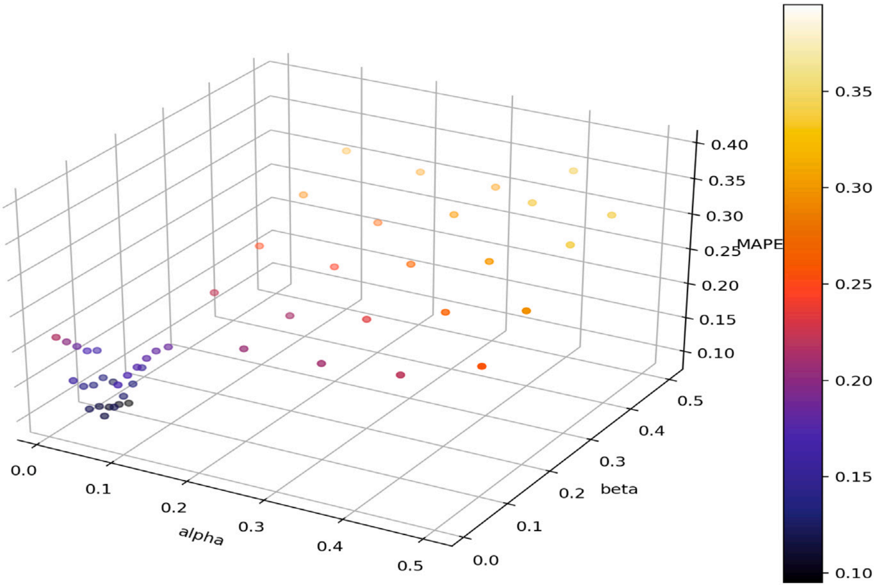

Fig. 9. The effect of model parameters on MAPE.

- Phase I: The sensitivity analysis is performed for a wide variety of parameter values within the interval [0, 1]. Results reveal that α and β values greater than 0.5 result in a high error when predicting the total migratory flux. In addition, the results suggest that the optimal values for α and β must be between 0 and 0.25 (see Fig. 8).
- Phase II: After Phase I results, it can be concluded that the optimal parameter values should be smaller than 0.5. Therefore, the grid search was conducted for combinations of α and β values ranging from 0.1 to 0.5 with 0.1 increments. The migration data from 1970 to 1985 was utilized in the calibration procedure in order to preserve data for the validation procedure. The MAPE metric was used to evaluate the results. The optimal MAPE values were obtained when alpha and beta were equal to 0.1. Fig. 9 depicts the MAPE value for the sought-after parameter values. It indicates that the optimal values for both parameters are less than 0.1.
- Phase III: The grid search was conducted for combinations of α and β values ranging from 0.01 to 0.09 with 0.2 increments in this phase. The best MAPE values were 0.09518 with alpha equal to 0.05 and beta equal to 0.07. With decreasing parameter values, the MAPE value worsens. Therefore, it can be concluded that optimal parameter values for α are near to 0.05 and for β are close to 0.07, and the obtained parameter values are near to optimal. We stopped searching for better model parameters because the MAPE value was at an appropriate level and due to the time required to perform each experiment. Please note that all experiments in the calibration process are conducted with the same seed number so that the randomness of the model is eliminated and the effect of parameter values may be observed clearly.

Until this point, simulation experiments are conducted and computer simulations are verified against the conceptual model. Thus, Steps 3 and 4 of the framework are implicitly applied.

4.6. Model validation

This section represents the final phase of the framework. For the validation of the model, macro-level data and a valid model from the literature are also utilized. After calibrating the model parameters, we compared the theory-supported ML-driven ABM results with observed values and theory-driven ABM findings. The overall migratory flux predicted by theory-driven ABM and theory-supported ML-driven ABM is depicted in Fig. 10. We computed the MAPE for each methodologies in order to more objectively compare their performance. It is 10.11% for theory-driven ABM and 8.81% for theory-supported ML-driven ABM. We have calculated these MAPE values for the years 1986 to 1999. In terms of MAPE, both models yield reasonably decent results. The outcomes of theory-supported ML-driven models are superior to those of theory-driven models because theory-supported ML-driven models make use of data while constructing models.

Both theory-driven and ML-driven ABM fail to capture the observed values for the years 1980 to 1986. We believe that the failure of the models in Burkina Faso could be a result of unusual and more country-level occurrences, such as environmental difficulties, economic policies of the country, political issues, etc.

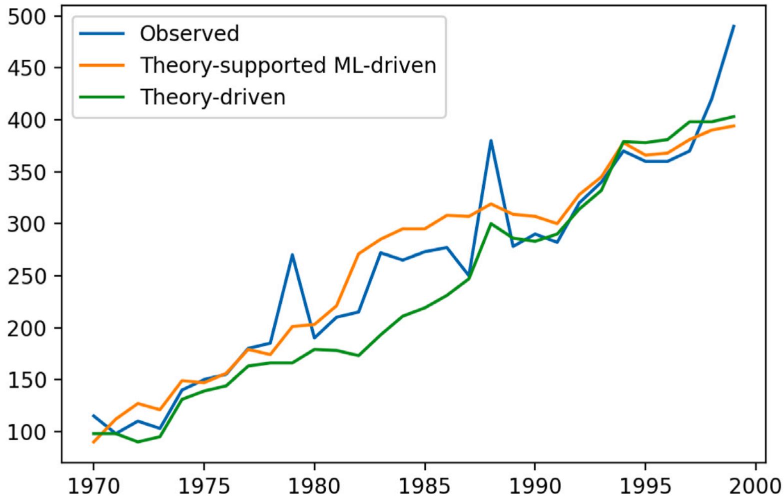

Fig. 10. Comparison of theory-supported ML-driven ABM and theory-driven ABM.

5. Challenges/discussions

The proposed framework might guide the researchers in the development of ML-driven ABMs, particularly in the modeling of agent behaviors. We have demonstrated the applicability of the framework through a case study and compared its outcomes to those of the theory-driven version of the model. Considering the accuracy of the model, the outcomes are encouraging.

On the other hand, the framework is limited by its data requirements. As in the traditional ABM building procedure, macro-level data is required to ensure the calibration and validation of the model in models built using the proposed framework. Individual-level data is also needed to model the behaviors of agents in the proposed framework. It may not be simple for the modeler to obtain both individual-level and macro-level data. With today's technology, it is possible to collect data about an individual via a mobile phone app or a data tracking system. However, this data may require processing before it can be used to simulate individual-level behavior. Moreover, the success of built models is contingent upon the quality of the data. Without high-quality data, the modeler cannot construct effective ABMs, as the performance of the ML model trained with poor data will be inadequate. In a nutshell, the modeler faces the following data-related obstacles and requirements while constructing a model using the framework:(i) obtaining appropriate and high-quality data, and (ii) transforming data into a form that can be used.

The framework recommends searching the literature and consulting experts before collecting individual-level data for modeling the behavior of agents. So the modeler can avoid collecting useless data and instead collect the right data to predict agent behavior with high accuracy. This may also help the modeler in preprocessing and feature selection if individual-level data is readily available.

The framework suggests incorporating the interaction between agents during model construction; otherwise, The model will be missing, i.e., it will not be a totally ABM. However, the data set may not have a feature for including the effect of interaction when modeling agent behavior using ML. In such situations, the framework suggests combining the interaction factor with the output of the ML model. However, there is no exact way to combine them. In our example study, we multiply the ML output by the interaction impact. It may be performed in several different ways. The modeler must decide how to do so by ensuring that the effect of interaction is successfully incorporated into their model. This is an another challenge that a modeler may encounter when constructing an ABM utilizing the framework.

Another debate concerning the framework revolves around which ML algorithms should be employed. The framework suggests using white-box models if the modeler wants to interpret the predicted behavior of agents. Due to their focus on explaining the relationship between variables, interpretable ML approaches may be also capable of fulfilling this need as stated in the study by Molnar [81]. If, on the other hand, the modeler is only interested in predicting macro-level output with a high degree of accuracy and is unconcerned with the interpretability of the macro-level output of ABM, the modeler may use readily available individual data and black-box ML models. This research does not make use of the interpretability capability of the white-box models because of the study objectives. Individual migration preferences may be linked to the features utilized in ML models if white-box models are studied more thoroughly.

Another issue of debate concerning the developed model is its generalizability. Because developing a model in this manner may be interpreted as making it more data dependent. However, we believe that the generalizability of all ABMs remains a concern because their calibration and validation are performed on a limited data set. Using literature-supported features to train an ML model may also improve its generalizability.

The proposed framework can be utilized for different ABMs with slight changes. A modeler might create ML-driven ABMs based on the suggested framework by gathering the requisite individual data for modeling agent behavior and macro-level data for calibrating and validating the model. In further, having the real-world data regarding the simulation environment and agent's attributes for the initialization of the simulation model could enhance the model's reliability and accuracy.

6. Conclusion

Today, there is an ever-increasing amount of data being generated and stored. Using these data to create ABMs could potentially facilitate the development of these models. It could also improve the accuracy of ABMs' predictions. This study attempts to explicate the relationship between ABMs and ML techniques and how and why ML models may be incorporated into ABMs. The particular emphasis is placed on the application of ML models for simulating agent behaviors in the models.

A main scheme is proposed for directing researchers who want to employ ML models for simulating agent behavior in their ABMs. The main scheme is broken down into three sub-frameworks. One of these sub-frameworks is explained and illustrated using a real case study. The results of the case study support the usefulness of the provided framework for constructing ABMs. To demonstrate the generalizability of the suggested framework, however, it must be applied to further real-world examples. The future research objective is to examine the applicability of the proposed framework to further case studies and enhance its generalizability based on the results of these case studies.

Two sub-frameworks of the main scheme have yet to be discussed in detail or illustrated with a real-world example. These frameworks should be developed and shown in real case studies in order to fully guide researchers in employing ML approaches to represent agents' behavior in constructing models. A further objective of future study is to develop these sub-frameworks and provide a comprehensive scheme for enhancing the applicability of ML techniques in ABMs and the accuracy of ABMs.

Data availability

The authors do not have permission to share data.

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